Research Article Vol. 34, No. 10 / 18 May 2026 / Optics Express 18676
Optics EXPRESS

# Unrolled Richardson-Lucy deconvolution network with partially connected layers in computational microscopy

FENG TIAN AND WEIJIAN YANG*

*Department of Electrical and Computer Engineering, University of California, Davis, Davis, CA 95616, USA*
*wejyang@ucdavis.edu

**Abstract:** Computational microscopy systems often employ optical encoders with non-localized point spread functions (PSFs) to enable advanced imaging capabilities. A notable example is the multi-cluster PSF, commonly used in light-field microscopes and mask-based integrated microscopes for single-shot 3D imaging. These systems rely on computational decoders to reconstruct high-quality images from complex optical measurements. Traditional iterative optimization methods, while physically interpretable, are computationally intensive. In contrast, deep neural networks offer faster inference but often lack physical interpretability. Physics-informed neural networks aim to bridge this gap by integrating model-based insights into data-driven architectures. Here, we propose an unrolled neural network architecture [partially connected unrolled Richardson-Lucy (RL) network (PC-RLN)], which is inspired by the RL deconvolution algorithm, and tailored for microscopy with non-localized, multi-cluster, and spatial variant PSFs. Each stage of the network mimics an iteration of the RL algorithm, and incorporates a learnable, partially connected layer to model both the forward imaging process and the back-projection. This design enables efficient and interpretable image reconstruction with substantially reduced computational cost compared to the standard RL deconvolution algorithm. We demonstrate the effectiveness of our approach in both Fourier light-field microscopy and mask-based integrated microscopy with microlens array.

© 2026 Optica Publishing Group under the terms of the Optica Open Access Publishing Agreement

## 1. Introduction

Conventional optical microscopy typically employs a localized point spread function (PSF), and the captured images often closely resemble the object. In contrast, computational microscopy [1] often designs optical encoders with complex, non-localized PSFs to enable advanced functionalities, such as single-shot 3D imaging [2–4]. These PSFs are often spatially extended and sometimes spatially variant. A notable class of such PSFs consists of spatially distributed discrete points or clusters, which can encode object depth [5–18]. Representative systems using these multi-cluster PSFs include mask-based integrated microscopes with microlens arrays (MLA) [6–8,12,13,15,16] and Fourier light-field microscopes [5,9,14]. In both cases, each object point is imaged by multiple lenslets, thus resulting in the non-localized, multi-cluster PSF structure. In the mask-based integrated microscopes, the PSF is often spatial variant.

To reconstruct the object from measurements, computational decoders are required. The reconstruction algorithms generally fall into two categories: model-based iterative optimization and deep neural networks. Model-based methods explicitly incorporate the forward imaging model (and thus the PSF), and aim to fit the object to the measurement through iterative refinement. A well-known example is the Richardson-Lucy (RL) deconvolution algorithm, which is an iterative maximum-likelihood method assuming Poisson noise in measurement. RL alternates between forward projection of the current estimate and back-projection of the residual using the PSF [4]. While effective, RL often amplifies noise in later iterations, leading

#588318 https://doi.org/10.1364/OE.588318
Journal © 2026 Received 23 Dec 2025; revised 13 Mar 2026; accepted 22 Apr 2026; published 13 May 2026

Optics EXPRESS

to ill-convergence [19]. Other iterative methods incorporate regularization which is often based on object sparsity, and are solved using techniques such as gradient descent or alternating direction method of multipliers (ADMM) [20]. The iterative optimization methods are generally computationally expensive, especially for non-localized and spatial variant PSFs where the entire system matrix needs to be stored.

Learnable approaches, on the other hand, construct networks models to directly map measurements to object reconstructions [21]. These methods typically require large datasets of image-object pairs for training. Once trained, they offer fast inference. Many such networks use generic architectures like U-Net [22], which, while powerful, may lack generalizability and physical interpretability. Recently, physics-informed neural networks have emerged, which embed domain-specific knowledge into the network design [10–13,15–17,23–27]. For example, in Fourier light-field microscopy, images are decomposed into multiple views and then merged via a neural network [25–27]. In mask-based imaging systems, deconvolution is performed in the Fourier domain, where shallow networks are employed to learn optimized Fourier filters [11,12,15,16]. Furthermore, specialized architectures have been developed to specifically address the challenges of spatially variant point spread functions (PSFs) [15,16,24,28–30].

A particularly promising direction is the development of unrolled neural networks, which bridge model-based iterative algorithms with deep learning [16,23,31–34]. These networks start from a known iterative algorithm (e.g., RL, ADMM) and unroll its operations into a trainable architecture with learnable parameters. This yields a highly interpretable workflow, where each network stage corresponds to an iteration of the original algorithm. One example is to unroll the ADMM algorithm [23] or primal-dual optimization framework [33]. However, it requires substantial computational resources when the PSF is spatial variant. Another example is to unroll the RL algorithm, represented by RL network (RLN) [34]. RLN embeds the physics of image formation and the update rule into a trainable structure, allowing the forward and back-projection parameters to be learned through convolution layers. Compared to conventional black-box networks, RLN uses fewer parameters, achieves faster processing, and generalizes better to unseen samples, especially along the challenging axial dimension. However, RLN is primarily designed for systems with localized PSFs and requires substantially more computation resources for systems with non-localized, multi-cluster PSFs.

In this work, we develop a new unrolled RL network tailored for systems such as integrated microscopes with microlens array masks and Fourier light-field microscopes, which feature non-localized, multi-cluster PSFs. Instead of using convolutional layers to model the forward and backward projections as in RLN [34], we employ partially connected layers, which are better suited for modeling non-localized, multi-cluster PSFs. Additionally, our architecture is designed to accommodate spatially variant PSFs. We term this architecture Partially Connected Unrolled RL Network (PC-RLN). Depending on the loss function, the network can be trained in either a supervised or self-supervised manner. We demonstrate the effectiveness of our approach on two applications: Fourier light-field microscopy and MLA-integrated microscopy. Our network achieves 2-3 orders of magnitude faster processing speed and uses 4-6 orders of magnitude less RAM than conventional RL algorithms, while maintaining high reconstruction accuracy.

## 2. Partially connected unrolled RL network

### 2.1. Background of RL deconvolution

RL deconvolution algorithm iteratively refines an estimated object $\hat{o}(x)$ from the measured image $i(x)$ with a known PSF. The object is updated as below at the $k^{\text{th}}$ iteration

$$\hat{o}^{k+1}(x) = \hat{o}^{k}(x) \cdot \left[ \frac{i(x)}{\hat{o}^{k}(x) \otimes PSF(x)} \otimes \widetilde{PSF}(x) \right], \tag{1}$$

Optics EXPRESS

where $\otimes$ represents convolution and $\widetilde{PSF}$ represents the mirrored $PSF$. Each iteration essentially consists of four steps. (1) Forward projection: the current object estimate is projected to the image domain via convolution with the PSF; (2) Error computation: the ratio between the measured image and the forward-projected image is computed through pixel-wise division, which serve as the error signal; (3) Backward projection: the error is mapped back to the object domain via convolution with the mirrored PSF, to create a correction signal; (4) Object update: the object estimate is updated by element-wise multiplication with the correction signal.

While the RL algorithm is computationally efficient for systems with localized PSFs, it becomes more demanding when the PSF is spatially extended or spatially variant. In the former case when the PSF is spatially extended, the memory and computational load increase in step (1) and (3) due to the larger PSF footprint. To reduce the computational resources, step (1) and (3) could be implemented in Fourier domain, but it is subject to high frequency noise of the PSF. In the latter case when the PSF is spatially variant, a full system matrix is required to describe the spatially varying PSF, which dramatically increases complexity. While the PSF may be decomposed to multiple spatial invariant PSFs using singular value decomposition, which allows the image to be synthesized from the object without constructing the system matrix, this process still incurs substantial computational cost.

To address this, we previously developed a list-RL algorithm tailored for MLA-integrated microscopy [16], where the PSF is spatially variant and composed of multiple discrete clusters. Each cluster, corresponding to the PSF of a lenslet, was modeled as a single point. Two mapping lists were constructed: one for forward projection (object voxel to image pixel) and one for backward projection (image pixel to object voxel). Equation (1) can be cast as below

$$\hat{o}^{k+1} = \hat{o}^{k} \cdot \boldsymbol{B}\left(\frac{i}{\boldsymbol{F}(\hat{o}^{k},\ PSF)}, PSF\right), \tag{2}$$

where $\boldsymbol{F}$ represents voxel-to-pixel mappings and $\boldsymbol{B}$ represents the pixel-to-voxel mappings. This list-based approach significantly reduces the computational cost by avoiding the need for a full system matrix. However, it simplifies the PSF model by modeling each cluster as a single point, and fixing the weights among different points, which could lead to suboptimal reconstructions.

### 2.2. Architecture of partially connected unrolled RL network

The PC-RLN proposed in this work builds upon the list-RL algorithm while addressing its limitations. PC-RLN is a cascade of $N$ iterative deconvolution stages that explicitly mimic the operations of the RL algorithm (Eq. (2)), but with trainable components. Each stage corresponds to one iteration of an RL update, consisting of a learnable forward projection, an error computation, a learnable backward projection and object update (Fig. 1). The forward projection and backward project is realized by a partially connected layer to model the voxel-to-pixel mapping and pixel-to-voxel mapping. Here, the full PSF is not stored and only effective mappings between pixel-voxels are used to build the model connections, and their weights are learned. A convolutional-layer-based regularization module is incorporated after the forward and backward projection to enhance stability, suppress noise and enhance resolution. Compared to the classical RL algorithm, PC-RLN substantially reduces computational requirements and improves reconstruction performance and robustness. Compared to the list-RL, PC-RLN offers enhanced flexibility and accuracy by learning optimal weights and incorporating regularization.

The forward projection module is implemented as a partially connected layer, which maps individual object voxels to image pixels. This layer is constructed by thresholding the designed PSF: if a PSF value exceeds the threshold, a connection is established between the corresponding voxel and pixel. While this thresholding simplifies the model and reduces computational complexity, it inevitably omits some weaker connections. To compensate for this loss, the weights of the established connections are learnable, allowing the network to adaptively adjust the forward

Optics EXPRESS

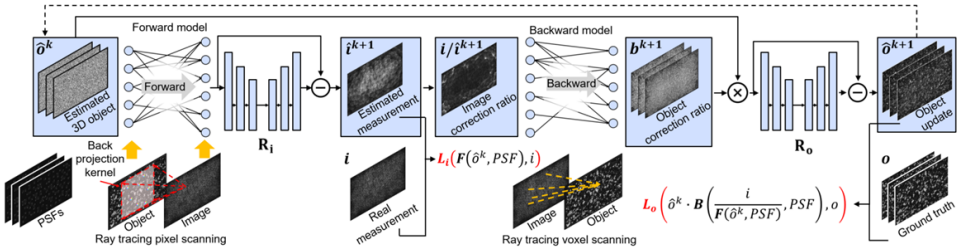



**Fig. 1.** Basic architecture of the partially connected unrolled RL network (at the k$^{th}$ iteration). The unrolled RL network operates iteratively, with each cycle refining the object estimate. The PSF comprises multiple spots. The initial object guess is generated as a 3D stack of random noise. A partially connected network, initialized using the PSF spot locations, models the forward imaging process by projecting object voxels onto image pixels. The resulting estimated image is compared with actual measurements to compute an image correction ratio. This ratio is then propagated through a second partially connected network that models the backward imaging process, which maps image pixels back to object space and produces an object correction ratio. The object correction ratio guides the update of the object estimate for the next iteration. To enhance stability and accuracy, a convolutional-layer-based regularization module is applied after both forward and backward projections. The loss function $\mathcal{L}_o$ and $\mathcal{L}_i$ measures the difference between the updated object and the ground truth, and the forward model output and the real measurements, respectively.

model during training. These learnable weights play a crucial role in recovering the influence of the pruned connections, enabling the network to maintain reconstruction fidelity despite the sparsified structure.

The backward projection module follows the same strategy, mapping the error signal from the image domain back to the object domain, based on the pixel-voxel connection established by PSF structures.

Both of the regularization modules right after the forward and backward projection module are implemented through regularization-by-denoising (RED) [35,36] with a small convolutional neural network (CNN), typically starting from $11 \times 11$ kernel size and reducing to $3 \times 3$ with tanh activation. These modules learn the noise and artifacts, which can be subtracted from the data. Different from the hand-crafted regularization functions such as TV, the regularization here is data-driven and adaptive.

With the regularizer $\mathbf{R_i}$ after the forward projection and $\mathbf{R_o}$ after the backward projection, the image and object update at each unrolled/iteration stage can be expressed as:

$$\hat{i}^{k+1} = \boldsymbol{F}(\hat{o}^k,\ PSF) - \mathbf{R_i}(\boldsymbol{F}(\hat{o}^k,\ PSF)), \tag{3}$$

$$\hat{o}^{k+1} = \hat{o}^k \cdot b^{k+1} - \mathbf{R_o}(\hat{o}^k \cdot b^{k+1}), \tag{4}$$

where $b^{k+1}$ is object correction ratio signal produced by the trained backward partially connected layer. All operations, including multiplications, convolutions, and updates, are fully differentiable, enabling an end-to-end training of the entire unrolled pipeline. All the unrolled or iteration stages use the same defined forward / backward module and regularization module.

The number of unrolled or iteration stages, $N$, was determined empirically by balancing reconstruction performance against computational cost. Our experiments indicated $N = 3$ yields near-optimal deconvolution quality, with further iterations offering diminishing returns.

The major reconstruction module in PC-RLN is physical projection through partially connected layers, making it extremely lightweight compared to typical CNNs used in image reconstruction, which often contain large data flow [37]. For systems with non-local PSFs or spatial variant PSF,

Optics EXPRESS

our partially connected PC-RLN architecture avoids redundant pixel-voxel mappings inherent in full system matrices used in conventional RL deconvolution. In the example of Fourier light-field microscopy and MLA-integrated microscopy shown in Section 3, our method uses only $10^{-4}$~$10^{-6}$ amount of connections compared to that of the fully connected reconstruction layers to store the PSF information, resulting in substantial speed improvements and memory savings. This compact architecture enables efficient processing of large-scale 3D volumes.

### 2.3. Loss functions and network training

We design two primary loss functions in PC-RLN: $\mathcal{L}_o$ for the reconstructed object volume which compares the reconstruction to the object ground truth, and $\mathcal{L}_i$ for the predicted measurement which compares the forward model output to the real measured image (Fig. 1).

$$\mathcal{L}_{o_2D} = \lambda_{o_2D} BCE(o,\ \hat{o}), \tag{5}$$

$$\mathcal{L}_{o_3D} = \lambda_{o_3D}(SSIM(o_{xy},\ \hat{o}_{xy}) + SSIM(o_{xz},\ \hat{o}_{xz}) + SSIM(o_{yz},\ \hat{o}_{yz})), \tag{6}$$

$$\mathcal{L}_i = \lambda_i BCE(i,\ \hat{i}), \tag{7}$$

where $\hat{o}$ and $\hat{i}$ denote the final reconstructed object volume and the predicted image respectively. $o_{\mathrm{xy}}$, $o_{\mathrm{xz}}$, $o_{\mathrm{yz}}$ and $\hat{o}_{\mathrm{xy}}$, $\hat{o}_{\mathrm{xz}}$, $\hat{o}_{\mathrm{yz}}$ denote the XY, XZ, YZ projection views, respectively, of the ground truth object volume and the reconstructed object volume. For 2D sample, we use binary cross-entropy (BCE) for $\mathcal{L}_o$, as it promotes pixel values and structural fidelity. For 3D volumetric objects, we use structure similarity index measure (SSIM) computed from orthogonal projection views for $\mathcal{L}_o$, which can enhance the axial resolution and thus volumetric reconstruction quality compared to BCE (Supplement 1 Fig. S1). We set weight $\lambda_{o_2D} = 1.0$, $\lambda_{o_3D} = 1.0$, and $\lambda_i = 1.0$. Here, $\mathcal{L}_i$, $\mathcal{L}_o$ can be used independently or in combination. This flexible loss design enables both supervised and self-supervised learning strategy. Self-supervised training uses only $\mathcal{L}_i$, requiring no ground truth object volumes. Supervised training uses $\mathcal{L}_o$, optionally combined with $\mathcal{L}_i$, when ground truth volumes are available.

Training data can be generated either through simulation or experimental acquisition. The PSFs used in training can be either synthesized using a wave-optics model or measured directly from the microscope system.

## 3. Results

### 3.1. Fourier light-field microscopy

Fourier light-field microscopy [5,9] is a special type of light-field microscopy where the microlens array is at the Fourier plane of the objective to record the 4D light field (Fig. 2(a)). The PSF appears to contain multiple clusters (Fig. 2(b)), each being generated by a lenslet. The separations of the clusters depend on the object depth. As an example, we configured 4 lenslets in the Fourier plane. We evaluated the effectiveness of PC-RLN to recover the 3D object volume from the Fourier light-field images through simulated data.

We generated synthetic 3D volumes ($160 \times 160 \times 7$ voxels) containing structures such as microtubule networks, beads, and nuclei arrangements, which was adopted from [25]. We used wave optics beam propagation methods to synthesize light field images (Fig. 2(c)), adapted from [25] with added Poisson noise. Here, the system PSF was approximated by a shift-invariant 3D PSF (Fig. 2(a)-(b)) [25]. This enables the use of standard convolution operations to synthesize the datasets.

We trained the network using the simulated dataset. The PSF is used to initialize the partially connected layer. We first set a threshold to remove the background noise and low intensity lobes in PSF spots, then index the rest pixels in the PSF as valid mapping pixels that encode object information. There could be multiple pixels kept for each cluster in the PSF. These pixel-voxel

Optics EXPRESS

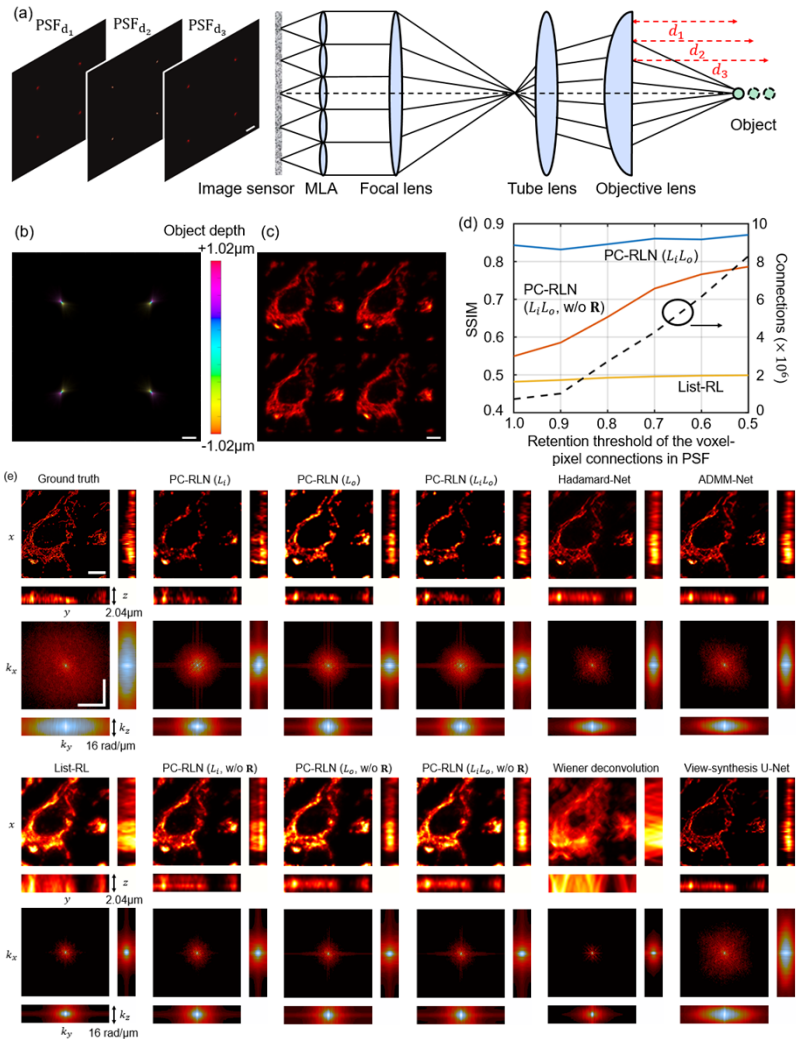



**Fig. 2.** 3D reconstruction of Fourier light field microscopy using synthetic dataset. (a) Schematics of the Fourier light field microscopy, and illustration of the depth dependent PSFs. (b) Overlaid PSFs from different depths. (c) Raw measurement of a 3D sample. (d) Left axis: SSIM of the reconstruction from PC-RLN (trained with $L_i$ and $L_o$; with or without regularizers **R**) versus list-RL at different retention threshold of the voxel-pixel connections in PSF. 1.0: only the most significant PSF pixel in each cluster is kept; 0.9-0.5: the PSF pixels above 90%-50% of the max PSF intensity in each depth are kept. Right axis: the number of connections at different retention threshold of the voxel-pixel connections in PSF. (e) Comparison of the 3D reconstruction among different methods. PC-RLN were trained with different loss functions ($L_i$ or $L_o$, or both $L_i$ and $L_o$; with and without regularizers **R**). The object contains 7 depths. Top, reconstruction results in spatial domain, presented in $xy$, $yz$ and $xz$ view. Bottom, the spatial frequency domain spectrum, presented in $k_xk_y$, $k_yk_z$ and $k_xk_z$ view. The reconstruction results from PC-RLN and list-RL were median filtered with $3 \times 3$ kernel to remove voxel artifacts. The reconstruction results from Hadamard-Net and ADMM-Net were normalized at each depth to remove depth-dependent bias. Scale bar, (a)-(c), 500 μm. (e), 5 μm (spatial domain); 1000 rad/μm (spatial frequency domain).

Optics EXPRESS

maps will be used to build the architecture of partially connected layer in forward and backward projections. Training was performed using the Adam optimizer with a learning rate of 0.001 for 50 epochs over 60 training datasets. Each dataset has an object of $160 \times 160 \times 7$ voxels and an image of $384 \times 384$ pixels. Thanks to its lightweight architecture, PC-RLN can be trained efficiently (Supplement 1 Fig. S2), which required ~3 GB RAM on a single NVIDIA A10G GPU with float 32 (for real number) and complex 64 (for complex number) training precision. Inference is fast: deconvolving a volume with $160 \times 160 \times 7$ voxels takes ~27 ms, compared to tens of seconds or over a minute for 3 iterations of standard RL (depending on whether CPU or GPU is used). This speedup is attributed to both the reduced number of iterations and the efficiency of the partially connected layers.

We evaluated the performance of 3D reconstruction of mitochondria samples [25] using different loss functions in PC-RLN: $\mathcal{L}_i$ only, $\mathcal{L}_o$ only, and both $\mathcal{L}_i$ and $\mathcal{L}_o$ combined (Fig. 2(e), Table 1, Supplement 1 Fig. S3). The supervised learning model trained with both $\mathcal{L}_i$ and $\mathcal{L}_o$ achieved the best performance, indicated by a higher SSIM and multi-scale SSIM (MS-SSIM), as well as a higher peak signal-to-noise ratio (PSNR) and lower mean squared error (MSE) of the reconstruction against the ground truth, and a sharper resolution. Supervised training with $\mathcal{L}_o$ only tended to over smooth the object, which may cause the forward model deviate from system property and thus degrade the reconstruction performance. Interestingly, the self-supervised learning model trained with only $\mathcal{L}_i$ performed very well, better than the supervised learning model trained with $\mathcal{L}_o$. To investigate the significance of the convolutional regularizers, we removed them both after the forward projection and backward projection module in an ablation study. Notably, without the regularizers, the reconstruction performance dropped, indicated by the poor MSE, SSIM, MS-SSIM and PSNR, and the spatial frequency spectrum range (Fig. 2(e), Table 1). Through the convolution regularizers, the reconstructed pixel/voxels could be better connected to its nearby pixels/voxels, which extend the capability of the partially connected layers.

We further investigated how the number of voxel-pixel connections within each PSF cluster influences the reconstruction performance, by varying the retention threshold of the connections in the designed PSF (Fig. 2(d)). Interestingly, PC-RLN (trained with both $\mathcal{L}_i$ and $\mathcal{L}_o$) performed well even with only one connection within each PSF cluster, and maintained stable performance with only slight improvement as the number of connections increases. In contrast, PC-RLN without regularizers performed worse in all cases, and depended more heavily on a larger number of connections to achieve good results. This again reiterates the significance and functionality of the convolution regularizers: they enable a reduced number of connections in the partially connected layer by efficiently linking reconstructed pixels/voxels to their neighbors through convolution.

We benchmarked PC-RLN against list-RL [16] (Fig. 2(d)-(e), Table 1, Supplement 1 Fig. S3). List-RL resembles RL but utilizes a simplified and fixed forward and backward projections. For a fair comparison, we modified the original list-RL algorithm [16] to allow multiple voxel-pixel connections per PSF cluster. With the same number of connections, list-RL underperformed compared to PC-RLN (with and without regularizers) (Fig. 2(d)). These results emphasize the critical role of learning connection weights in PC-RLN (Supplement 1 Fig. S4).

Finally, we also benchmarked PC-RLN against a range of classical and state-of-the-art methods, including Wiener deconvolution, Hadamard-Net [12], ADMM-Net [16] and view-synthesis U-Net [26] (Fig. 2(e), Table 1, Supplement 1 Fig. S3). Wiener deconvolution represents a classical non-iterative baseline; as a linear filter with fixed regularization, it yielded the lowest performance. In contrast, Hadamard-Net, ADMM-Net, and the view-synthesis U-Net are sophisticated neural network: Hadamard-Net excels learning in the Fourier domain for PSFs with large frequency support, ADMM-Net utilizes the unrolled ADMM optimization framework, and the view synthesis U-Net is tailored for Fourier light-field reconstruction. To minimize

Optics EXPRESS

**Table 1. Quantitative metrics of different reconstruction methods for Fourier light-field microscopy, using dataset from [25]. The results were averaged from a total of 8 sample volumes. SSIM, MS-SSIM and PSNR were calculated at individual depth and reported as depth-averaged number; MSE were calculated volumetrically. The parameter RAM refers to the static memory, i.e., the RAM storing the model weights. The activation RAM refers to the dynamic memory, i.e., the RAM storing activation/feature maps produced by the convolution channels. The formulation to calculate the parameter RAM and activation RAM is described in Supplement 1 Table S1.**

| Method / Metric | SSIM | MS-SSIM | PSNR (dB) | MSE | Inference time (ms) | Parameter RAM (MB) | Activation RAM (MB) | Stored connections / PSF kernel size | Fully connected system matrix size |
|---|---|---|---|---|---|---|---|---|---|
| PC-RLN (trained with $L_i$) | **0.8498** | **0.8532** | **29.1327** | **0.0025** | 27 | 20.7498 | 13.2227 | $2.713\times10^6$ | $2.642\times10^{10}$ |
| PC-RLN (trained with $L_o$) | 0.8459 | 0.8439 | 27.2400 | 0.0048 | | | | | |
| PC-RLN (trained with $L_i$ and $L_o$) | **0.8579** | **0.8610** | **29.5796** | **0.0029** | | | | | |
| PC-RLN (trained with $L_i$, w/o regularizers **R**) | 0.6065 | 0.6919 | 23.3030 | 0.0072 | 27 | 20.7032 | 2.4922 | $2.713\times10^6$ | |
| PC-RLN (trained with $L_o$, w/o regularizers **R**) | 0.5985 | 0.6888 | 22.5479 | 0.0093 | | | | | |
| PC-RLN (trained with $L_i$ and $L_o$, w/o regularizers **R**) | 0.6054 | 0.6949 | 23.1180 | 0.0092 | | | | | |
| List-RL | 0.4925 | 0.5830 | 17.3404 | 0.0190 | 27 | 20.7032 | 2.4922 | $2.713\times10^6$ | |
| Weiner deconvolution | 0.1474 | 0.2545 | 14.6716 | 0.0348 | 23 | 3.9375 | 16.4336 | $1.032\times10^6$ | |
| Hadamard-Net | 0.6718 | 0.7757 | 31.5567 | 0.0018 | 20 | 7.8750 | 16.4336 | | |
| ADMM-Net (6 stages) | **0.8425** | **0.8707** | **32.9406** | **0.0016** | 31 | 11.7542 | 85.4766 | | |
| View synthesis U-Net | **0.8710** | **0.9008** | **35.2103** | **0.0011** | 21 | 0.9784 | 28.8574 | None | |

costs, the ADMM-Net employs a strategy that bypasses the large-scale matrix multiplications used in [16]. While these three networks, all designed for the spatially invariant PSFs used here, outperformed PC-RLN in some of the quantitative metrics, PC-RLN maintained a competitive margin. Crucially, a distinct advantage of PC-RLN is its support for self-supervised learning, a capability not shared by the other three architectures. In terms of computational resources, the inference times and the memory usage (Supplement 1 Table S1) were overall comparable across the methods (Table 1). Crucially, while the standard RL algorithm requires system matrix storage that scales with the product of the voxel and pixel counts, PC-RLN and list-RL scale linearly with the number of voxels (weighted by the connectivity factor). This puts the computational resources of our proposed PC-RLN in the same order of magnitude as methods that do not store the system matrix such as Hadamard-Net, ADMM-Net (modified to bypassing the large-scale matrix multiplication as noted above) and view-synthesis U-Net, highlighting its memory efficiency. Compared to these advanced neural networks, PC-RLN requires more static memory (parameter RAM) to store the parameter/weights, but less dynamic memory (activation RAM) to store the activation/feature maps.

Optics EXPRESS

### 3.2. MLA-integrated microscopy

Integrated microscopy is an emerging imaging technique that utilizes a single layer of mask as an optical encoder [2,3,38,39] (Fig. 3(a)). This approach enables a compact device footprint while achieving a large FOV across a 3D volume. Among various mask designs, MLAs are particularly effective, as they produce sparse PSFs that are well-suited for imaging densely featured objects [6–8,12,13,15,16]. The resulting PSF consists of multiple clusters, each corresponding to the PSF of an individual lenslet. Here, we evaluate the performance of the PC-RLN reconstruction framework for objects imaged by an MLA-integrated microscope, for both synthetic and experimental datasets.

We generated the training data using the depth-dependent PSF calculated from random microlens array pattern, and projected 3D objects onto image through each of the microlens units. The object contains 13 depths of randomly generated alphabet letter shapes. The system has an optical resolution of ~10 μm with a magnification of ~0.35. On the imaging plane, the sampling rate is 1.85 μm/pixel, which satisfies the Nyquist sampling criterion for the specified resolution. We initialized and trained PC-RLN using a similar strategy as that in Fourier light field microscopy in Sec. 3.1. Training was performed using the Adam optimizer with a learning rate of 0.001 for 50 epochs over 40 training datasets. Each dataset has an object of $256 \times 384 \times 13$ voxels and an image of $256 \times 384$ pixels. Training was performed on NVIDIA Tesla A10G GPU with 24 GB RAM. Inference time for each test set is ~94 ms.

We evaluated the trained PC-RLN using datasets consisting of randomly distributed beads (Fig. 3(b)) and alphabet letter shapes (Fig. 3(c)). While PC-RLN was trained using alphabet letter shapes, it performed well in both types of samples. Similar as in the Fourier light-field microscopy in Sec 3.1, the performance of PC-RLN was benchmarked against list-RL [16] and Hadamard-Net [12], ADMM-Net [16] and view-synthesis U-Net [26] (Fig. 3(b)-(c), Table 2). Overall, PC-RLN demonstrated superior performance in both visual fidelity and quantitative metrics. Because the PSF in this system is spatially variant, and Hadamard-Net, ADMM-Net and view-synthesis U-Net utilized a single PSF kernel, they suffered from degraded reconstruction and lateral feature shifts (Supplement 1 Fig. S5) toward the edges of the field of view (FOV). In contrast, PC-RLN maintained high reconstruction accuracy across the entire FOV. Furthermore, PC-RLN achieved superior axial resolution, characterized by a significantly reduced axial span of reconstructed features. Within the PC-RLN variants, the model trained with both $\mathcal{L}_i$ and $\mathcal{L}_o$ yielded the best reconstruction quality overall.

We further applied PC-RLN to an experimental dataset captured by an MLA-integrated microscope, as reported in [16], where experimentally acquired images served as training dataset (Fig. 4, Table 3). The optical resolution and system magnification are the same as described above. The samples consisted of fluorescently coated alphabet letters randomly distributed on a photomask, and the ground truth images were obtained via a benchtop fluorescence microscope. To simulate volumetric data imaged by the MLA-integrated microscope, we captured images at varying sample-to-microscope distances and then summed images from different depths. The raw measurements consisted of $2048 \times 3072$ pixels, while each depth of the reconstruction had $595 \times 907$ pixels. Due to GPU memory constraints, the PC-RLN model was implemented using the design pattern of the microlens array with a single-pixel approximation for the PSF of each lenslet. While this approximation is effective under minimal aberration (as shown in Fig. 2(d)), it introduces a performance trade-off due to the aberration in the lenslet from the imprecise fabrication. Including more pixels in the PSF could improve the performance here. Despite this computational simplification, the experimental results mirrored our findings in the synthetic datasets: PC-RLN consistently outperformed list-RL, Hadamard-Net, ADMM-Net, and the view-synthesis U-Net.

Optics EXPRESS

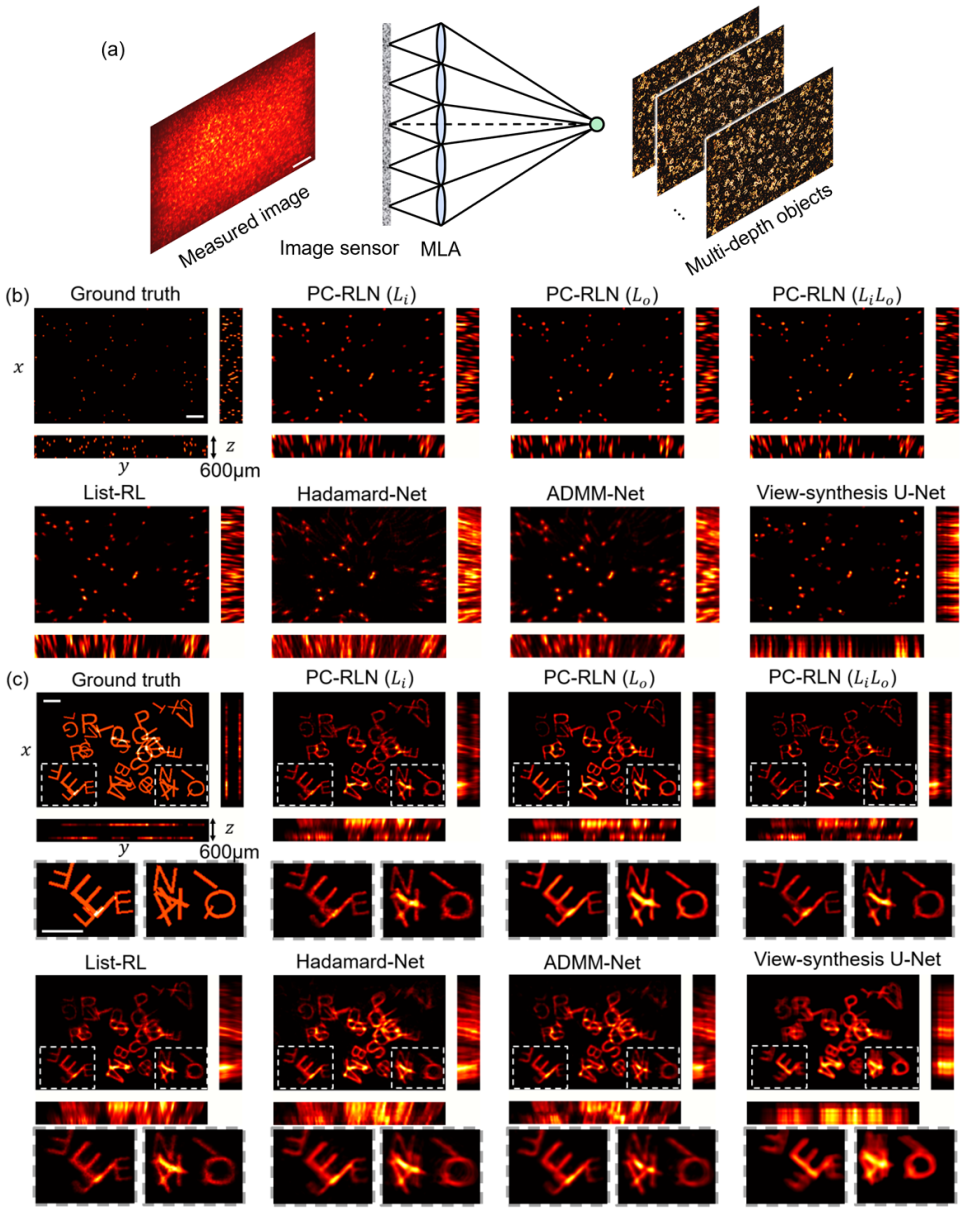



**Fig. 3.** 3D reconstruction of MLA-integrated microscope using synthetic dataset. (a) Schematics of the MLA-integrated microscopy. (b) Comparison of the 3D reconstruction of a sample containing randomly distributed beads, presented in xy, yz and xz view, for different reconstruction methods. PC-RLN were trained with different loss functions ($L_i$ or $L_o$, or both $L_i$ and $L_o$). The object contains 13 depths. (c) Same as (b), but for a sample containing letters distributed in different depths. Zoomed-in views were shown for two sub fields of view. The reconstruction results from all methods were median filtered with $3 \times 3 \times 3$ kernel to remove voxel artifacts and mean-subtracted. The reconstruction results from Hadamard-Net and ADMM-Net were normalized at each depth to remove depth-dependent bias. Scale bar, (a)-(c), 200 μm; (c) inset, 200 μm.

Optics EXPRESS

**Table 2. Quantitative metrics of different deconvolution methods for MLA integrated microscope, using simulated dataset from [16]. The results were averaged from a total of 6 sample volumes including alphabet letter samples. SSIM, MS-SSIM and PSNR were calculated at individual depth and reported as depth-averaged number; MSE were calculated volumetrically.**

| Method / Metric | SSIM | MS-SSIM | PSNR (dB) | MSE | Inference time (ms) | Parameter RAM (MB) | Activation RAM (MB) | Stored connections / PSF kernel size | Fully connected system matrix size |
|---|---|---|---|---|---|---|---|---|---|
| PC-RLN (trained with $L_i$) | 0.8319 | 0.7855 | **23.2366** | 0.0109 | 94 | 142.0237 | 47.2500 | $5.611\times10^6$ | $1.256\times10^{11}$ |
| PC-RLN (trained with $L_o$) | 0.8459 | 0.7856 | 21.0228 | 0.0138 | | | | | |
| PC-RLN (trained with $L_i$ and $L_o$) | **0.8799** | **0.8241** | 23.2107 | **0.0106** | | | | | |
| List-RL | 0.7465 | 0.7081 | 19.9137 | 0.0134 | 94 | 142.0200 | 10.5000 | $5.611\times10^6$ | |
| Hadamard-Net | 0.6747 | 0.6571 | 21.1751 | 0.0121 | 35 | 87.7500 | 181.6699 | $1.278\times10^6$ | |
| ADMM-Net (6 stages) | 0.7579 | 0.7337 | 22.7181 | 0.0110 | 50 | 100.7230 | 804.8320 | | |
| View-synthesis U-Net | 0.6874 | 0.6570 | 18.5402 | 0.0203 | 26 | 0.9863 | 154.4854 | None | |

**Table 3. Quantitative metrics of different deconvolution methods for MLA imager, using experiment dataset collected from [16]. The results were averaged from a total of 3 test pairs. SSIM, MS-SSIM and PSNR were calculated at individual depth and reported as depth-averaged number; MSE were calculated volumetrically.**

| Method / Metric | SSIM | MS-SSIM | PSNR (dB) | MSE | Inference time (ms) | Parameter RAM (MB) | Activation RAM (MB) | Stored connections / PSF kernel size | Fully connected system matrix size |
|---|---|---|---|---|---|---|---|---|---|
| PC-RLN (trained with $L_i$) | 0.4220 | 0.3464 | 11.7813 | 0.0669 | 110 | 91.4547 | 754.2422 | $1.198\times10^7$ | $6.791\times10^{12}$ |
| PC-RLN (trained with $L_o$) | 0.4155 | **0.3514** | **11.9353** | **0.0646** | | | | | |
| PC-RLN (trained with $L_i$ and $L_o$) | **0.4240** | 0.3327 | 11.4817 | 0.0716 | | | | | |
| List-RL | 0.4040 | 0.3392 | 11.7570 | 0.0673 | 110 | 91.4547 | 59.3438 | $1.198\times10^7$ | |
| Hadamard-Net | 0.2741 | 0.2458 | 11.5781 | 0.0699 | 31 | 96.0000 | 197.6719 | $1.258\times10^7$ | |
| ADMM-Net (6 stages) | 0.3290 | 0.2770 | 11.4735 | 0.0715 | 47 | 110.7183 | 1292.3438 | | |
| View-synthesis U-Net | 0.1928 | 0.1487 | 10.3511 | 0.0927 | 159 | 1.02927 | 1172.6601 | None | |

Optics EXPRESS

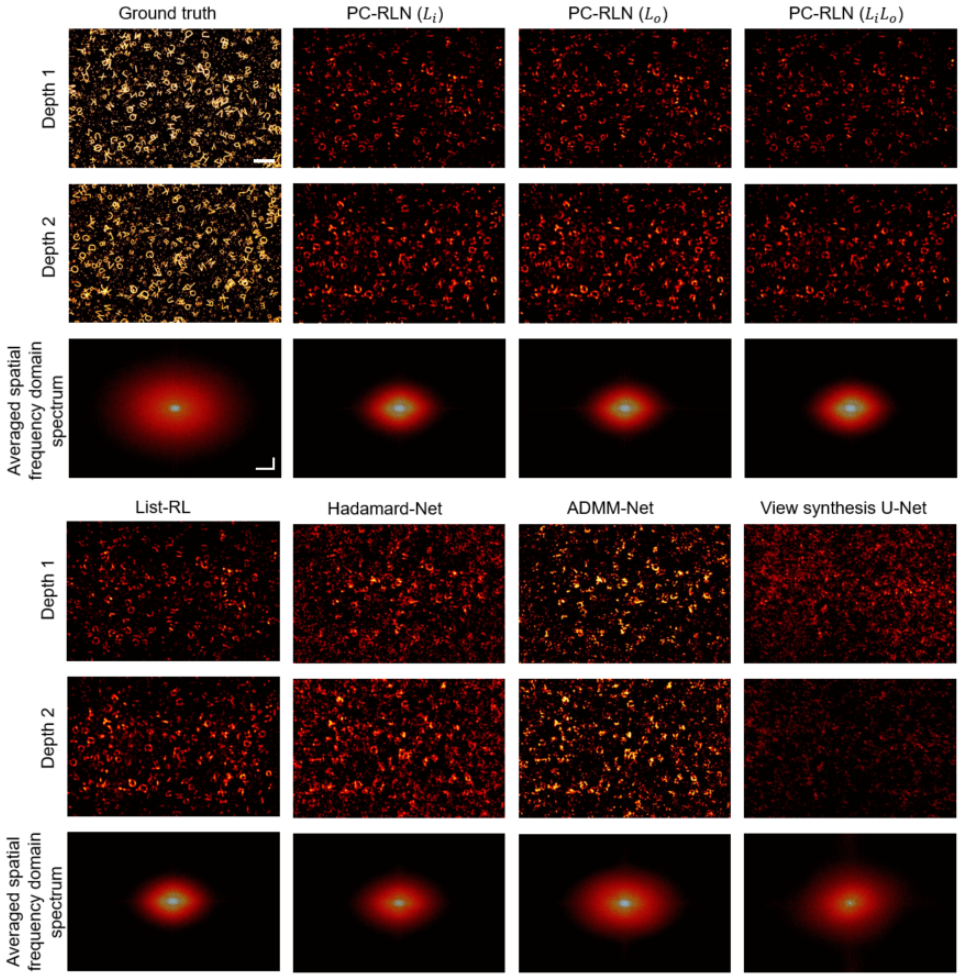



**Fig. 4.** 3D reconstruction of MLA-integrated microscope using experimental dataset, which composed two layers/depths of randomly distributed alphabet letters. PC-RLN were trained with different loss functions ($L_i$ or $L_o$, or both $L_i$ and $L_o$). Top, reconstruction results in spatial domain for depth 1; middle, reconstruction results in spatial domain for depth 2. Bottom, the averaged spatial frequency domain spectrum. The reconstruction results from PC-RLN and list-RL were median filtered with $5 \times 5$ kernel to remove voxel artifacts. For all the methods, the reconstruction results were mean-subtracted. Scale bar, 500 μm (spatial domain), 0.1 rad/μm (spatial frequency domain).

## 4. Summary and discussion

We present PC-RLN, a light-weight, learnable unrolled RL deconvolution network designed for real-time computational microscopy with multi-cluster PSFs. In these scenarios, the PSF is distributed as multiple clusters across the entire FOV, making fully convolutional or fully connected networks computationally inefficient.

PC-RLN builds upon the RL iterative scheme by unrolling its iterations into a trainable network with physically interpretable operations. Unlike the previously published Deep-URL [32] or RLN [34] which use convolutional networks to learn the forward or backward projections and thus has small receptive field, PC-RLN addresses PSFs that span large FOVs by modeling the

Optics EXPRESS

forward and backward projection process through partially connected layers that map the image and object spaces. Crucially, the connections in these layers are sparse. Compared to the fully connected layers, PC-RLN significantly reduces the computational cost, while retaining a good reconstruction quality. Compared to list-RL, PC-RLN allows adjusting the weights among the connections. PC-RLN further incorporated CNN-based regularization modules which locally refine the image or object volume for denoising and edge sharpening.

We validated PC-RLN in Fourier light-field microscopy and MLA-integrated microscope, both of which encode 3D object information into a single image using MLAs. Compared to advanced architectures such as Hadamard-Net [12], ADMM-Net [16] and notably the view-synthesis U-Net [26], which is the backbone of the state-of-the-art deep learning reconstruction algorithm designed for Fourier light field microscopy (e.g. F-VCD [25], DFLFM [26] and HFLFM [27]), PC-RLN maintains highly competitive performance. While it exhibits a margin in reconstruction quality, PC-RLN offers a distinct advantage: by integrating data-driven learning with model-based interpretability, it reduces the reliance on extensive ground-truth datasets and supports a self-supervised learning workflow.

In MLA-integrated microscope where the PSF is spatial variant, PC-RLN is particularly well-suited. It outperformed architectures that rely on a single, spatial invariant PSF kernel such as the Hadamard-Net [11, 12], and ADMM-Net [16]. While SV-FourierNet [15], multi-Hardmard-Net [16] and multi-ADMM-Net [16] were developed to account for spatially variant PSFs, they require more complex architectures and higher computational resource. In contrast, PC-RLN offers a simpler, shallow-layer design while having the capability for self-supervised learning.

In summary, PC-RLN transforms the conventional RL algorithm into a learnable unrolled architecture that balances physical model fidelity with data efficiency. PC-RLN is computationally efficient, interpretable, support self-supervised learning, and is particularly well-suited for 3D microscopy applications where the PSFs form multiple clusters across the FOV. This makes PC-RLN a promising tool for real-time, high-resolution volumetric imaging in biological research.

**Funding.** National Institute of Neurological Disorders and Stroke (R01NS133924); Burroughs Wellcome Fund (CASI 1015761).

**Disclosures.** The authors declare no conflicts of interest.

**Data availability.** Data underlying the results presented in this paper and the code of the reconstruction algorithms are available from [40].

**Supplemental document.** See Supplement 1 for supporting content.

## References

1. J. N. Mait, G. W. Euliss, and R. A. Athale, "Computational imaging," Adv. Opt. Photonics **10**(2), 409–483 (2018).
2. J. K. Adams, V. Boominathan, B. W. Avants, *et al.*, "Single-frame 3D fluorescence microscopy with ultraminiature lensless FlatScope," Sci. Adv. **3**(12), e1701548 (2017).
3. N. Antipa, G. Kuo, R. Heckel, *et al.*, "DiffuserCam: lensless single-exposure 3D imaging," Optica **5**(1), 1–9 (2018).
4. M. Broxton, L. Grosenick, S. Yang, *et al.*, "Wave optics theory and 3-D deconvolution for the light field microscope," Opt. Express **21**(21), 25418–25439 (2013).
5. C. Guo, W. Liu, X. Hua, *et al.*, "Fourier light-field microscopy," Opt. Express **27**(18), 25573–25594 (2019).
6. Y. Xue, I. G. Davison, D. A. Boas, *et al.*, "Single-shot 3D wide-field fluorescence imaging with a Computational Miniature Mesoscope," Sci. Adv. **6**(43), eabb7508 (2020).
7. K. Yanny, N. Antipa, W. Liberti, *et al.*, "Miniscope3D: optimized single-shot miniature 3D fluorescence microscopy," Light: Sci. Appl. **9**(1), 171 (2020).
8. F. Tian, J. Hu, and W. Yang, "GEOMScope: Large Field-of-view 3D Lensless Microscopy with Low Computational Complexity," Laser Photonics Rev. **15**(8), 2100072 (2021).
9. X. Hua, W. Liu, and S. Jia, "High-resolution Fourier light-field microscopy for volumetric multi-color live-cell imaging," Optica **8**(5), 614–620 (2021).
10. S. S. Khan, V. Sundar, V. Boominathan, *et al.*, "FlatNet: Towards Photorealistic Scene Reconstruction from Lensless Measurements," IEEE Trans. Pattern Anal. Mach. Intell. **44**(4), 1934–1948 (2022).
11. D. Deb, Z. F. Jiao, R. R. Sims, *et al.*, "FourierNets enable the design of highly non-local optical encoders for computational imaging," *Advances in Neural Information Processing Systems* 35 25224–25236 (2022).
12. F. Tian and W. Yang, "Learned lensless 3D camera," Opt. Express **30**(19), 34479–34496 (2022).
13. Y. Xue, Q. Yang, G. Hu, *et al.*, "Deep-learning-augmented computational miniature mesoscope," Optica **9**(9), 1009–1021 (2022).

Optics EXPRESS


14. R. Guo, Q. Yang, A. S. Chang, *et al.*, "EventLFM: event camera integrated Fourier light field microscopy for ultrafast 3D imaging," Light: Sci. Appl. **13**(1), 144 (2024).
15. Q. Yang, R. Guo, G. Hu, *et al.*, "Wide-field, high-resolution reconstruction in computational multi-aperture miniscope using a Fourier neural network," Optica **11**(6), 860–871 (2024).
16. F. Tian, B. Mattison, and W. J. Yang, "DeepInMiniscope: Deep learning-powered physics-informed integrated miniscope," Sci. Adv. **11**(37), eadr6687 (2025).
17. Y. Y. Li, Z. X. Zhang, F. Tian, *et al.*, "V-shaped PSF for 3D imaging over an extended depth of field in wide-field microscopy," Opt. Lett. **50**(2), 383–386 (2025).
18. L. A. Kabuli, H. Pinkard, E. Markley, *et al.*, "Designing lensless imaging systems to maximize information capture," Optica **13**(2), 227–235 (2026).
19. Y. M. Liu, S. Panezai, Y. T. Wang, *et al.*, "Noise amplification and ill-convergence of Richardson-Lucy deconvolution," Nat. Commun. **16**(1), 911 (2025).
20. S. P. Boyd, *Distributed optimization and statistical learning via the alternating direction method of multipliers* (Now Publishers Inc., 2011).
21. B. Zhu, J. Z. Liu, S. F. Cauley, *et al.*, "Image reconstruction by domain-transform manifold learning," Nature **555**(7697), 487–492 (2018).
22. O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, N. Navab, J. Hornegger, W. M. Wells, and A. F. Frangi, eds. (Springer International Publishing, 2015), pp. 234–241.
23. K. Monakhova, J. Yurtsever, G. Kuo, *et al.*, "Learned reconstructions for practical mask-based lensless imaging," Opt. Express **27**(20), 28075–28090 (2019).
24. K. Yanny, K. Monakhova, R. W. Shuai, *et al.*, "Deep learning for fast spatially varying deconvolution," Optica **9**(1), 96–99 (2022).
25. C. Q. Yi, L. X. Zhu, J. H. Sun, *et al.*, "Video-rate 3D imaging of living cells using Fourier view-channel-depth light field microscopy," Commun. Biol. **6**(1), 1259 (2023).
26. X. Hua, W. Liu, and S. Jia, "High-resolution Volumetric Cell Imaging with Deep Fourier Light-Field Microscopy (DFLFM)," in *Optica Imaging Congress (3D, COSI, DH, FLatOptics, IS, pcAOP)* (Optica Publishing Group, 2023), paper DM1A.3.
27. Y. Huang, X. Zhao, and D. Li, "Hybrid Fourier light field microscopy system with deep learning for 3D high-resolution reconstruction," Opt. Express **33**(23), 47545–47562 (2025).
28. S. Howard, P. Norreys, and A. Döpp, "CoordGate: Efficiently Computing Spatially-Varying Convolutions in Convolutional Neural Networks," arXiv (2024).
29. J. W. Wei, Y. H. Liu, J. C. Wu, *et al.*, "Thin and lightweight camera based on Pancake lens and deep learning," Opt. Lett. **49**(17), 4851–4854 (2024).
30. Q. Yang, Z. Chen, J. Zhang, *et al.*, "Coordinate-conditioned Deconvolution for Scalable Spatially Varying High-Throughput Imaging," arXiv (2026).
31. H.-C. Ko, J.-Y. Chang, and J.-J. Ding, "Deep Priors Inside an Unrolled and Adaptive Deconvolution Model," presented at *Computer Vision: 15th Asian Conference on Computer Vision* (2020), Revised Selected Papers, Part II.
32. C. Agarwal, S. Khobahi, A. Bose, *et al.*, "Deep-Url: A Model-Aware Approach to Blind Deconvolution Based on Deep Unfolded Richardson-Lucy Network," *International Conference on Image Processing*, 3299–3303 (IEEE, 2020).
33. O. Kingshott, N. Antipa, E. Bostan, *et al.*, "Unrolled primal-dual networks for lensless cameras," Opt. Express **30**(26), 46324–46335 (2022).
34. Y. Li, Y. J. Su, M. Guo, *et al.*, "Incorporating the image formation process into deep learning improves network performance," Nat. Methods **19**(11), 1427–1437 (2022).
35. Y. Romano, M. Elad, and P. Milanfar, "The Little Engine That Could: Regularization by Denoising (RED)," SIAM J. Imaging Sci. **10**(4), 1804–1844 (2017).
36. E. T. Reehorst and P. Schniter, "Regularization by Denoising: Clarifications and New Interpretations," IEEE Trans. Comput. Imaging **5**(1), 52–67 (2019).
37. C. Qiao, D. Li, Y. Guo, *et al.*, "Evaluation and development of deep neural networks for image super-resolution in optical microscopy," Nat. Methods **18**(2), 194–202 (2021).
38. M. S. Asif, A. Ayremlou, A. Sankaranarayanan, *et al.*, "FlatCam: Thin, Lensless Cameras Using Coded Aperture and Computation," IEEE Trans. Comput. Imaging **3**(3), 384–397 (2017).
39. V. Boominathan, J. T. Robinson, L. Waller, *et al.*, "Recent advances in lensless imaging," Optica **9**(1), 1–16 (2022).
40. Yang Research Laboratory, "Partially-Connected-Unrolled-RL-Network (PC-RLN)," GitHub (2026), https://github.com/Yang-Research-Laboratory/Partially-Connected-Unrolled-RL-Network.